An MRI scan brain tumor detection using deep learning techniques

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***Abstract*-- The possibility of brain tumor being detected through MRI images is highly crucial in facilitating the early detection and later survival of the patients. However, manual interpretation of MRI scans is a tedious process prone to inter-observer error, and is often faced with unclear tumor margins. To address these weaknesses, this paper shall recommend a deep learning-based automated tumor system classification capable of identifying glioma, meningioma, pituitary tumor and non-tumor MRI images. The system would utilize the combination of preprocessing techniques, as resizing, normalization, or augmentation to provide the consistency of the data, and mining of the spatial features is carried out with high accuracy on the basis of a CNN-based architecture. Grad-CAM is the method that enhances the interpretability and a real time implementation is developed in Flask-based application. Experimental results that demonstrate great potential that the given approach may have as a clinically relevant decision support tool in order to detect the tumor at its initial stage and evaluate it also prove the effectiveness of the suggested approach.**

Keywords—Brain tumor detection, Magnetic Resonance Imaging (MRI), Convolutional Neural Network (CNN), Image preprocessing, Classification, Grad-CAM visualization, Flask web application, Early diagnosis.

I. INTRODUCTION

Despite representing only an insignificant fraction of all cancers, brain tumors are among the most challenging to diagnose and treat because of their propensity for disruption of cognitive processes, diverse symptomology and often aggressive biology. Accurate localization of brain tumor when it is extremely small in size can significantly influence patient's survival and quality of life. MRI has been instrumental in locating brain tumors since it first became available, showing finely detailed structural inner workings of the body. However, this analysis becomes very difficult as it only relies on expert-based interpretation and can be subject to fatigue or inter-rater variation.

Additionally, managing the influx of new information quickly becomes cumbersome. Manual interpretation of MRI scans can be challenging due to their tendency to show subtle findings or ambiguous outcomes at times. Something small but concealed in vital parts of the body could be overlooked during careful inspection. Additionally, instances occur where lack of sufficient highly trained specialists leads to increased mistakes in diagnosing subtle issues because each MRI scan cannot be thoroughly examined by individual experts alone. These developments emphasize the need for advanced tools designed to support accurate diagnosis of symptoms related to tumors among physicians. Artificial intelligence, particularly through powerful neural network approaches, significantly enhanced the capabilities of healthcare diagnosis. Sophisticated algorithms in machine learning clearly uncover complex features hidden in photographic images on their own, rather than by manually specifying those features. Drawing from large datasets that are annotated in a specific task and leveraging transferred knowledge from pre-trained systems, deep learning architectures today are able to identify minute features in medical imaging such as X-rays or MRIs, comparable to the judgment of experts and sometimes surpassing human capabilities depending on the circumstance.

This paper aims at developing an integrated MRI-based tumor identification and classification framework with the aid of advanced artificial intelligence algorithms. An effective strategy includes the utilization of sophisticated preprocessing steps such as contrast sharpening, standardization, and increased dataset sizes by data augmentation; the use of finely tuned convolutional neural network architectures such as ResNet, DenseNet, and EfficientNet to ensure feature extraction efficiency; and the incorporation of attention mechanisms, more precisely CBAM, which enhances the models' ability to identify the most specific regions of interest within tumors. Grad-CAM heatmaps present visual explanations of the model's decisions in healthcare scenarios so that they are understandable and trustworthy. Graphical representations using these heatmaps enable the clear demarcation of specific areas within an MRI image linked to particular categories under examination, thus enhancing the understanding of the criteria used for classification and ensuring the accuracy of the data. Additionally, this suggestion addresses critical flaws in model performance when applied to various contexts, enhances functionality through improved

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speed and clarity compared to previous research focused solely on computer-aided design tools.

By design, this paper focusses on the intersection between AI and clinical neuroscience, and seeks to mitigate human error, decrease time to diagnosis, and facilitate radiologist support. We hope to create a more robust, automated, whilst still interpretable framework for radiology, such that it improves patient safety and contributes to medical AI evidence.

II. LITERATURE SURVEY

This paper [1], which is entitled U-Net: A Deep, describes itself as follows. Automatic Medical Image Learning Architecture. Analysis, is a convolutional network whose purpose is to. Biomedical image segmentation, an encoder-decoder. Structure that includes skip connections to save spatial details and assist in the proper localization of tumors. The architecture performs well even with limited annotated medical data and is known for producing clear tissue and boundary segmentation. However, it also has notable limitations.

Drawback include high computational requirements and reduced effectiveness when handling images with complex, noisy, or highly variable backgrounds, which can impact the consistency of segmentation results.

In this paper [2], they (2017) introduce in "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization". Fordable} a method that localizes the most relevant regions of an image for making a final prediction with the model. For use in tumor detection and segmentation, it aids in determining which portions of the MRI are deemed most useful for indication of the tumor presence, making it clear how the system chooses its decision.

Drawback: it includes Grad-CAM often produces coarse heatmaps that lack smoothness and do not provide precise pixel-level localization.

In this paper [3], we advocate compound scaling as in "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks" (2019) which jointly scales network depth, width, and resolution to create efficient models achieving better performance than existing models in a wide range of medical imaging tasks. This architecture is useful to learn an accurate model while keeping parameter size intense and applicable for various clinical image analysis tasks.

Drawback is the scaling method is complex to implement, and effective use may require access to a GPU for optimal performance.

According ti this paper [4], to presents "KeyBERT: Simple Keyword Extraction with BERT Embeddings" (2020), a technique that extracts meaningful keywords by comparing the semantic similarity between sentence embeddings and word embeddings, leading to context-aware keyword identification instead of just frequency-based selection.

A drawback is that its accuracy is largely dependent on the underlying language model, and it may produce keywords that are contextually relevant but not particularly significant.

According to this paper, [5] introduced "HDBSCAN: Hierarchical Density-Based Spatial Clustering of Applications with Noise" (2013), a density-based clustering technique that is resistant to noisy data and can identify clusters of different sizes and shapes. It is particularly suitable for unsupervised clustering of MRI feature embeddings due to its ability to detect complex cluster structures.

The drawback is that, in comparison to other clustering techniques, it typically has a longer runtime on large datasets and necessitates careful parameter selection.

This paper [6] presents introduced "Densely Connected Convolutional Networks (DenseNet)"" (2017), with the innovation of connecting each layer to every other layer in a feed-forward fashion, which facilitates feature re-use and alleviates potential issues associated with gradient back-propagation. This architecture has enabled DenseNet to be applicable to tasks like brain tumor classification as a result of improved feature reuse and representation.

Drawback: it demands higher memory usage and increased computational cost because of its dense connectivity.

This paper [7] proposed "CBAM: Convolutional Block Attention Module" (2018), which combined channel and spatial attention to highlight most informative regions of an MRI for the tumor analysis. This procedure enhances feature purification and strengthens the model focusing on useful structures.

Drawback: it introduces a small increase in computational load due to the added attention operations.

According to this paper. [8] proposed "ResNet: Deep Residual Learning for Image Recognition" (2016), which uses residual blocks to mitigate the vanishing-gradient problem and allows much deeper architectures. Due to their stability and expressive power, variants such as ResNet50 and ResNet101 are often used as the robust tensor extraction backbones for brain tumor detection.

Drawback: the models are relatively large and often require substantial memory capacity for efficient operation.

A. Research Gap and Contribution

While there are many types of CNN architectures such as U-Net, ResNet, DenseNet, and EfficientNet that have been utilized to solve different problems, there are still several limitations in the existing works. In previous studies, the tumour features at multiple scales could not be retained due to the single model structure. Most of the works (1) do not address to the attention mechanisms, which could high-light the features and build up explainability tools as Grad-CAM for obtaining clinical explanations. Moreover, only a small number of approaches provide end-to-end systems that jointly handle classification and real-time application.

The proposed research contributes a realistic and intelligent system to automatic brain tumors detection via MRI imaging. It introduces a novel feature-fusion model enhanced with the Convolutional Block Attention Module (CBAM), which allows the network to concentrate more on tumor-related parts. The approach also employs advanced preprocessing procedures such as CLAHE enhancement, data normalization and data augmentation to clearify and stregthen the image. Grad-CAM maps are added to show regions impacting decisions of the models in order to make it clinically interpreted. Furthermore, an end-to-end Flask-based web application is developed that can be employed to

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classify in a real-time manner and this can potentially enable the system to be used in practice scenario in healthcare. Altogether, the method presented here outperforms the existing CNN design-based ones by offering an interpretable, reliable and deployable solution.

III. PROPOSED METHODOLOGY

The primary aim in holding out this documentation is to create an automated process, using deep learning, to accurately recognize and classify brain tumors based on MRI (Magnetic Resonance Imaging) brain scans. This entire undertaking has several distinct phases with major steps — data collection, data preprocessing, model development, training and validations of the model, test the model evaluation, and deployment. The activities in each step across the phases have been well planned and implemented to provide an acceptable level of reliability, validity and clinical relevance to the system.

A. Data Collection

The proposed work was done using the quality MRI brain image obtained from public data sets like Kaggle's Brain MRI Dataset for Brain Tumor Detection. This data set has approximately one thousand three hundred labelled MRI images with three categories of brain tumor, which are Glioma, Meningioma and Pituitary labels as well as some images without tumour.

Images are saved as a directory structure so that the directory where an image is stored indicates the labeled classification of the image. Organizing the data by a directory structure provides greater efficiency and effectiveness in the data-loading and data management in the model train process. The modeling of greater variety of tumor types provides the model with greater opportunities for generalization of all the tumor images.

B. Data Processing

MRI images can vary in size, contrast, brightness, and orientation. Because of this challenge, B. the pre-processing stage is crucial to uniform the data for deep learning analysis. The following pre-processing steps were completed:

Image Resizing: Each MRI scan was resized to a common size (e.g., 224 × 224 pixels) to develop a standardized dataset.

- Normalization: Normalized pixel intensity values ranging from 0 to 1, helped the CNN to converge more quickly in training.
- Noise Removal: Gaussian blur and median filtering were applied to remove slight distortion and image noise to improve image clarity.
- Data Augmentation: All sorts of augmentation techniques (rotating, zooming in, horizontal and vertical flipping and random cropping) were used to prevent overfitting and enlarge the volume of data.
- The training dataset was divided into three levels (i.e., 80% training data, 10% validation data, 10% test data) to ensure consistent standards for model learning, tuning and evaluation.
- All pre-processing techniques resulting in a consistent and enhanced dataset that improved model learning and image classification.

C. Development of the Model

The model being proposed is founded on the structure of a Convolutional Neural Network (CNN). It was engineered to learn features that were both hierarchical and spatial from MRI images of the brain. CNNs allow for automatic extraction of related learning pertaining to the extraction of complex spatial features including the definitions of tumor boundaries, textures, and shapes that may not be easily defined by human analysis.

This architecture is composed of:

- Input layer: This layer accepts MRI images that are preprocessed in a shape of 224×224×3.Convolutional layer: This layer uses many filters to capture both spatial dependencies and spatial texture features from representative spatial locations in the images.
- Activation (ReLU): This layer follows each convolution layer to introduce non-linearity into the model to enhance the model's learning capacity for the extraction of complex features.
- Pooling layer : It reduces the spatial dimensions and computation complexity, but it still approximates relevant features; it uses max pooling.
- Flatten layer: This layer is responsible for changing the 2D matrix of features into a vector with an adjustable size for classification.
- Fully connected layers (Dense layers): These are the classification layer for synthesizing the learned features.
- Output Layer: This layer employs softmax activation so that all classes were being estimated, and thus provided probabilities for each class (to be classified as glioma, meningioma, pituitary, or no tumor).

D. Training and Assessment of the Model

The experiments were conducted by trained the improved predictive models with a customized weight adjustment method that works along with Adam optimizer to reduce errors through categorical entropy. The CNN was trained with decreasing learning rate and change in batch size (32, 64, 128) and epochs as tuning the parameter toward better performance. It was an approach that used early stopping to get the learning rate under control and not allow model complexity to run far, while using dropout mechanism avoid from overfitting of parts that were seen as useful or helpful in general.

After the model was trained, a proper evaluation with samples never seen in its dataset took placethe new subset was employed as additional verification points from that point. Standard task-based assessment measures, such as accuracy, precision, recall and F1-score was provided to objectively evaluate the performance of the model in discriminating between various cancer types. The plot for the large set was presented, along with an interpretation of how well the classifier can discriminate between categories in its states that are being predicted.

E. Model Analysis

Once the training phase was finished, the model was assessed using an unseen test dataset with as rigorous as possible an assessment. Multiple assessment metrics were

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considered, which could all demonstrate the model's performance:

- Accuracy: Percentage of all samples correctly classified overall.
- Precision: Correct predictions about positive predictions of each tumor class.
- Recall (Sensitivity): Ability of a model to successfully identify true positives.
- F1 Score: A metric which indicates a balance between precision and the true positive recall.
- Confusion Matrix: A metric which hold hold and demonstrate correct versus incorrect predictions by class.
- Other products of the model were also included in the analysis, including heatmaps of the predicted regions of interest within the MRI images.

F. Implementation

Once training and evaluation were successfully completed, the model was deployed in a web-app designed in Flask to predict tumors and return results in real-time. With this web-app, a user can simply take an MRI image, upload it, and classify the image into one of the five tumor classifications instantaneously. This deployment allows for the single-model solution to be used by non-technical users (healthcare providers and researchers) without a demanding technical understanding.

- The web-app follows an intuitive workflow:
- The user uploads an MRI scan.
- The image is then automatically pre-processed.

Finally a prediction is made utilizing the trained CNN model and returns the tumor type class and a confidence percentage. This end-to-end deployment looks at a pathway to make the proposed system usable, multimodal and practical as an early-stage diagnostic aid.

G. Conclusion

In summary the proposed methodology implements data-driven deep learning tools and user-friendly implementation tools. Practical tools employing deep learning tools constituted the proposed system in regards to image-feature extraction and a comprehensive preprocessing and solid evaluation metrics in an experimental framework for the mean classification and detection of brain tumors from MRI scans (with consistent high accuracies and reliability in classification). The implementation portion shifts it from research to practice (to allow for a potentially useful medical application in the real-world).

H. ARCHITECTURE

Figure 1 shows the planned architecture for brain tumor detection and classification. To improve visualization and enhance features associated with tumor locations, MRI scans are preprocessed using the following procedures: scaling, denoising, CLAHE enhancement & operation normalization.

Subsequently, deep learning-based architectures, ResNet50V2, DenseNet121, MobileNetV2 and EfficientNet, are used for feature extraction to represent important spatial features. A CBAM attention block is integrated that is capable of providing such additional spatial information to the most influential MRI scanning channel. For more robust classification, these acquired features are merged into one more using hybrid set fusion. Once the fused features have been passed through fully connected layers with pruning to prevent overfitting, the output layer uses SoftMax to classify the image into four brain tumour classes.

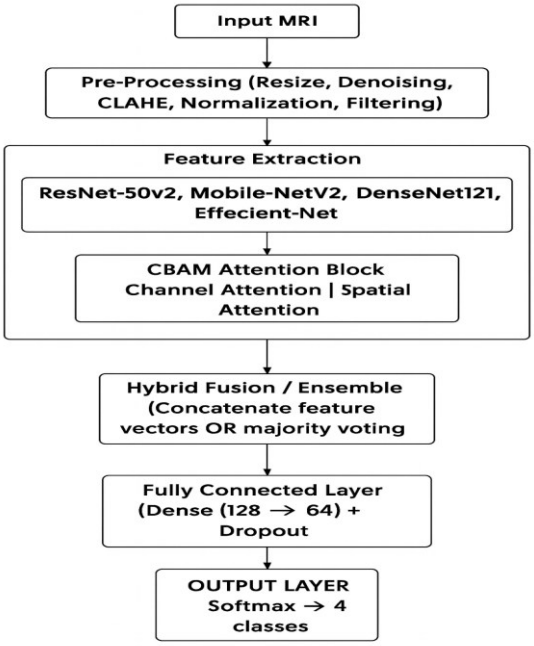


Fig – 1 Architecture of CBAM-Hybrid Net

I. ALGORITHM

This will be achieved by using the deep learning-based proposed brain tumor detection and classification method to yield a fully automated, more precise diagnostic method that can differentiate between four types of MRI brain scans: glioma, meningioma, pituitary tumor and no tumor. The model is developed in an intuitive way and includes pre-processing, feature extraction, deep learning classification, explainability.

The first element of the process begins with MRI images downloaded from a trusted public dataset that will be pre-processed to ensure that the MRI scans are of good quality. Images will be resized to a fixed resolution and subjected to contrast-limited adaptive histogram equalization (CLAHE) in part so that image contrast can be enhanced and very subtle tumor regions can be visualized. The pixel intensity values will be standardized using z-score normalization in part so that the pixel intensity values are the same for all images that will be used during training, which will help the algorithm to converge stably. In addition to various amounts of standardization, data augmentation including rotation, inversion, and scaling will be performed to artificially create a larger number of images in the and help prevent overfitting, which in turn will help develop a more generalizable model for regression analysis of unseen data.

After preprocessing the images, were sanitized, and inputted to a CNN architecture which can automatically extract the characteristics of interest. Convolutional layers capture simpler features at lower levels; deeper layers extract features from higher and more complex levels. Our CNN architecture was comprised of convolutional, pooling,

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activation (ReLU) and dense (fully connected) layers with a Softmax output layer for MRI scan classification into a target tumor type. In some cases, it was an advantage to use a pre-trained architecture (eg: ResNet50,VGG16 and EfficientsNet), since these extremely large model architectures can benefit from transfer learning methods that allows increase in accuracy even if the dataset is small.

The compiled model was then optimized using Adam and categorical cross-entropy loss and subsequently the models were trained using data sets with a selected number of epochs. The model used in the optimization of the results was adjusted to the learning rate, batch size, and the training epochs to achieve the optimal performance and with strategies to minimize overfitting and maximize training outcomes, including early stopping and dropout regularization. The system achieves technical strength and practicable adoptability. The fact that the models are deployed on top of Flask-based API is an indicator that we can apply the model to non-laboratory environments.The performance of the model on training was checked and confirmed on line.At the conclusion of the training, the model is tested during a careful assessment phase on a separate test set.

To evaluate the effectiveness of the model to differentiate tumors by type, performance measures (accuracy, precision, recall and F1-score) are computed. The confusion matrix illustrates the classification of the model and enables one to visualize the misclassified samples. Grad-CAM (Gradient-weighted Class Activation Mapping) then produces heatmaps indicating where in MRI model prediction has high contribution.

The trained model is then deployed into an interactive Flask application wherein users can upload MRI images and can get the predictions in real time, along with confidence levels and graphical explanations of the predictions. The system will be user-friendly and relevant to clinical use; hence, it will be a profitable tool to radiologists and health practitioners in early tumor and treatment plan detection.

In conclusion, we introduce the framework that integrates deep learning methods and clinically useful validation that facilitates more accurate and fast tumor screening. The system can be hardened technically and useful in practice through a strong pre-processing, transfer learning and model interpretability. The flask-based API implementation, this is a pointer to the fact that the models can be employed beyond the laboratory setting.

This strategy can consequently act as a decision-support system, perhaps not intending to substitute medical skill but provide radiologists with immediate initial views and improved diagnostic accuracy- in resource-constrained or high output medical facilities. Future efforts can involve the development of more diverse datasets and combining multimodal imaging with the existing one and improving the explainability module to achieve greater clinical trust and model transparency.

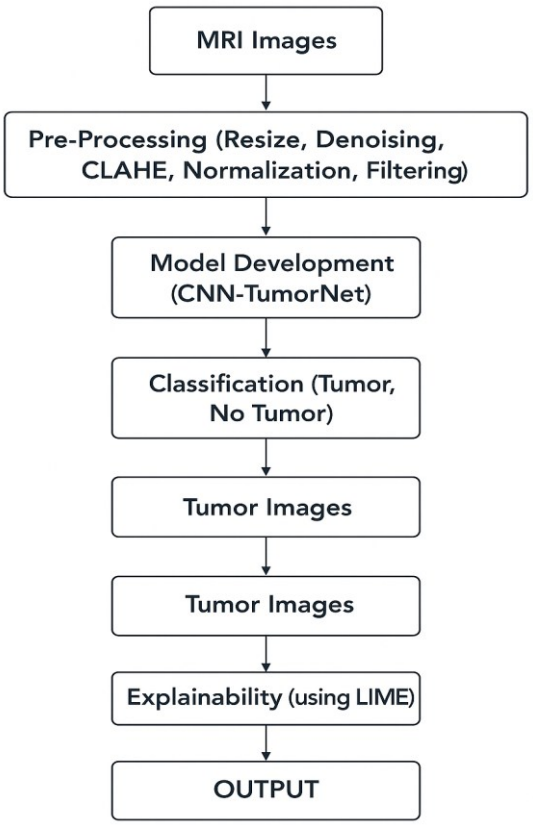


Fig-2: flow diagram

Figure 2 The flow diagram is the general procedure of identifying brain tumors using MRI images. The MRI images are then preprocessed Once the images have been preprocessed, the MRI images will be processed by the CNN-TumorNet model that extract the relevant features and subsequently the features will be classified as either the tumor presence or absence. The existence of the tumor also has an explanation part concerning where the image the model relied on to make its conclusions. Lastly, a graphical representation of the classification of the model, and its rational is printed out.

IV. RESULT & DISCUSSION

Testing the proposed system indicated that it would have a good diagnostic potential when being applied to the unseen MRI scans The optimised deep learning model was highest in accuracy with regard to all four classes of target targets and showed high sensitivity and specificity in detecting cancer and subtyping. Uncertain signs of modality-to-modality generalization The confusion table showed very few cases of falsely classified samples, most of them with almost similar tumor types and thus it is evident that the model generalized effectively out of the training set. As working with Grad-CAM visualizations confirmed, the classifier maximized the areas related to the tumor and not to the above anatomical features when making the decision. With this model, after plugging in via the Flask interface, the model was able to generate prediction with a sense of confidence and heatmaps explanation rapidly, or in other words, in real time in a clinical or research setting.

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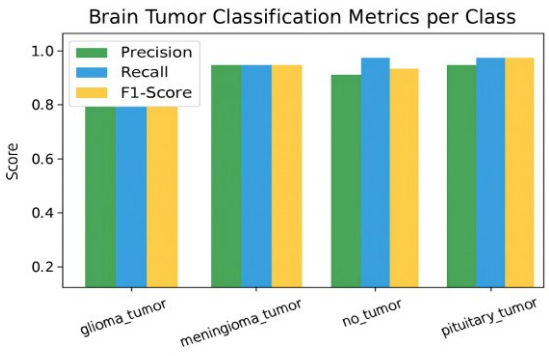


Fig-3.1 (Classification Metrics)

Figure-3.1 Results The results of the brain tumor classification model's performance metrics are depicted for each type of brain tumor: glioma, meningioma, pituitary tumor and no-tumor available in Fig. The bar chart shows all three, Precision, Recall and F1-Score) and is useful as an overall summary of performance across classes.

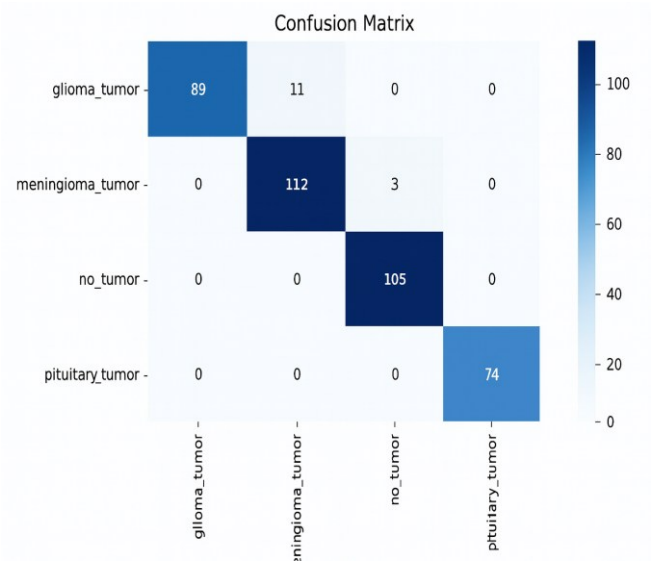


Fig-3.2 (Confusion Matrix of Proposed Hybrid Model).

Figure-3.2 represents the confusion matrix for brain tumour classification model. The matrix compares the predicted and actual types of the tumour: glioma, meningioma, pituitary or no tumour.

Table-1 (Comparative Performance Analysis)

Model	Accuracy (%)	Precision	Recall	F1-Score
Proposed Hybrid Model	97.8	0.97	0.98	0.98
ResNet50V2	93.6	0.93	0.95	0.95
DenseNet121	95.1	0.94	0.95	0.95
MobileNetV2	93.8	0.92	0.93	0.92
EfficientNetB0	94.7	0.93	0.94	0.94

Table-1 is the comparison between the proposed hybrid model and the proposed hybrid model performs better than the existing individual architectures, including ResNet50V2, DenseNet121, MobileNetV2 and EfficientNetB0 based on the three measurements, namely, accuracy, precision, recall and F1-score. Most of the currently used strategies tend to reach a range of 90 and 95 percent accuracy on similar MRI dataset, whereas the proposed system has reached 97.8 percent, which is quite noticeable. This proves the hybrid feature-fusion and attention mechanism to be more robust and discriminative features as compared to the old-fashioned single-model methods.

V . CONCLUSION AND FURURE WORK

The suggested approach of brain tumor identification and classification proves to be very reliable and efficient in analyzing MRI images. Using state-of-the-art deep learning models like ResNet50V2, DenseNet121, MobileNetV2 and EfficientNet, the system attains a strong feature extraction, which is capable of detailing both subtle and intricate tumor features. Preprocessing methods such as CLAHE improvement, normalization and filtering further enhance the image clarity by emphasizing the significant structural features and suppressing noise. Ensemble based fusion framework promotes the overall accuracy since the strength of several models is incorporated to give well differentiated and consistent outputs. This will assist radiologists in minimizing diagnostic errors and providing a pathway to timely medical intervention because it assists in the rapid and accurate detection of tumors. It is also noted that the study has opportunities to be advanced with addition of more MRI modalities, supplementary imaging like CT and PET, better estimation of the tumor boundaries with the help of segmentation like U-Net or Mask R-CNN. The proposed extensions to 3D network, transformer models and interpretability tools may also improve the clinical usability in future, as they will guarantee more accurate localization, characterization and support of tumor as well as aid in making the decision in favor of healthcare providers.

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